

CHASE NORVELL

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EDUCATION

San Diego Mesa College, San Diego, CA

- Mechanical Engineering | GPA: 3.95/4.00 | Dean's List | Planned Transfer to UC San Diego, Fall 2027
- Relevant Coursework: Linear Algebra, Properties of Materials, Circuits, Statics, Computer aided Design, Python Programming

EXPERIENCE

Mechanical Engineering Intern, Power Device Corp., San Diego, CA

Jun 2025 – Aug 2026

- Delivered 10+ PCB component-layout and manufacturing drawings with GD&T, supporting SolidWorks PCB and chassis assemblies and heat-transfer hardware.
- Evaluated thermal performance in ANSYS Mechanical and partnered with electrical engineers on design changes that reduced simulated peak temperature by ~16%.
- Automated ANSYS thermal post-processing using Python with AI-assisted debugging, cutting manual temperature extraction from ~1 hour to seconds.
- Designed and delivered 20+ production fixtures using SolidWorks, Fusion 360 CAM, 3D printing, and CNC machining, including one that cut setup time from ~5 minutes to under 1 minute.

Electronics Packaging Lead, Mesa Full Self Driving Club, San Diego, CA

Mar 2026 – Present

- Designed and delivered 6 custom electronics mounts for an autonomous kart, packaging PCBs, connectors, and power hardware for serviceable integration.
- Fabricated and integrated the electronics bay by cutting and sealing enclosure pass-throughs, routing power/signal cabling, and assembling mounting hardware, contributing to a Top 5 finish among 12 universities at Purdue's Autonomous Karting Series.

Mechanical Design / Team Lead, FIRST Robotics, San Diego, CA

Sept 2017 – Apr 2024

- Progressed over 7 years from mechanical member to design lead and Team Lead of a 15-person team, leading CAD, CDRs, prototyping, fabrication, and system integration.
- Redesigned scoring hardware with custom machined and 3D-printed components, roughly tripling throughput while reducing team scoring cycle time by ~86%.
- Designed drivetrain, intake, scoring, end-effector, and structural hardware across multiple robots, contributing to multiple design and engineering awards, including a Think Award at the 2024 FIRST Championship among 224 FTC teams worldwide.

SKILLS

- CAD & Analysis: SolidWorks, ANSYS Mechanical (FEA/Thermal), Fusion 360, Onshape, Autodesk Inventor
- Mechanical Design: GD&T, tolerance stack-ups, engineering drawings, DFM/DFA, interference & clearance analysis, Fixture / Tooling Design
- Manufacturing: Fusion 360 CAM, CNC milling, FDM/SLA 3D printing, soldering, silicone molding
- Programming & Electronics: Python, ANSYS automation, OrCAD/Allegro familiarity